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A new harvest time prognosis tool for forage maize production in Germany

Antje Herrmann *, Alois Kornher, Friedhelm Taube

Institute of Crop Science and Plant Breeding, Grass and Forage Science/Organic Agriculture, Christian-Albrechts-University, Olshausenstr. 40, D-24098 Kiel, Germany

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Abstract

Optimum harvest date of silage maize is crucial for exploiting the yield and forage quality potential of the crop, and for minimising losses during silage storage and the feedout phase. Simulation models are a useful tool for determining this optimal date. While numerous models are available for yield and some for quality, no comprehensive model deals simultaneously with yield and quality. The current project aims to fill this gap by (i) adapting the grass model FOPROQ to growth and quality of silage maize, and (ii) calibrating and validating the model for dry matter (DM) yield, DM content, and starch content. This allowed subsequently (iii) to perform a long-term simulation study for quantifying the climatic risk of silage maize grown under the marginal conditions of Northern Germany. Calibration was based on a field trial conducted at four sites over 3 years (1998–2000) in Germany, including six cultivars covering a wide range of maturity classes (early to mid-late) and maturity types (normal, dry-down, stay-green). Samples were taken every 2 weeks over the vegetation period. Validation was based on an independent data set collected during two years (2001, 2002) at two sites, however, with only one cultivar. No changes of algorithms were necessary to adapt the grass model to forage maize. Calibration results showed good agreement between observed and calculated data (RMSE values of 153.8 g DM m^{-2} for DM yield, 26.8 g DM kg^{-1}fresh weight for DM content, and 33.5 g kg^{-1} DM for starch content). For severe drought stress situations the validation revealed some discrepancies between observed and simulated starch content. Model refinements by implementing environmental response as a function of developmental stage and by broadening the calibration database are discussed. A 25-year (1979–2003) simulation study was conducted for one mid-early and two early cultivars at Kiel, Northern Germany. The simulation results were in good agreement with corresponding regional mean values for DM content and starch content of maize silage samples. A climatic risk analysis for this long-term simulation showed a substantial likelihood of failure to reach silage maturity, with failure rates between 44 and 64% depending on cultivar. The last decade (1994–2003), characterised by considerable temperature increase, showed somewhat lower failure rates. The reliable results presented in this study support current efforts to develop FOPROQ into a harvest time prognosis tool for Germany.
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Keywords: Forage maize; Modelling; Yield; Forage quality; Risk assessment

* Corresponding author. Tel.: +49 431 880 7420; fax: +49 431 880 4568.
E-mail address: aherrmann@email.uni-kiel.de (A. Herrmann).

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1. Introduction

The choice of cultivars as well as the timing of harvest should aim at maximising yield and quality of forage maize. In agricultural practice, however, forage maize is often harvested too early. This may cause severe problems of nitrogen in silage seepage effluent, especially in regions with unfavourable climatic conditions for maize growing such as Northwestern Europe. High ratios of maize silages with supra-optimal dry matter (DM) contents above 35% observed over the past years in Germany reveal another current problem, namely the risk of aerobic spoilage during the feedout phase (Ohyama et al., 1975). Improved prediction of forage maize maturity has become of increasing interest and is therefore extensively discussed at present.

During the last decades a number of models has been developed to simulate various aspects of yield formation and forage quality of maize. Process-oriented, dynamic models such as CERES Maize (Jones and Kiniry, 1986), MAIZE (Muchow et al., 1990), SUCROS97 (Van Laar et al., 1997), WOFOST (Boogard et al., 1998), APSIM (Carberry and Abrecht, 1991), CropSyst (Stöckle et al., 1994), and STICS (Brisson et al., 1998) focus mainly on maize grown for grain, which entails that DM content if at all is provided for grain only. The mathematical model of Stewart et al. (1997) simulates production, storage, and export of leaf sugar and starch, and was designed for research purposes rather than for practical application.

The current situation can therefore be characterised by the fact that none of the major models considers sufficiently many of the determining factors in order to successfully simulate yield and quality of forage maize. While dynamic, mechanistic models provide reliable simulations for DM yield and nitrogen concentration of the whole crop or of the grain, they do not include plant DM content and additional quality traits. Calculations based on thermal indices provide predictions of harvest time or forage quality traits (Dwyer et al., 1999; Easson and Fearnehough, 2003), but tend to disregard important environmental impacts others than temperature.

The dynamic FOPROQ (FOrage PROduction Quality) model was developed for grassland. It provides a comprehensive simulation of biomass production and of various forage quality parameters by considering the impact of environmental conditions, crop characteristics, and management inputs on yield formation and quality development (Torssell and Kornher, 1983; Kornher et al., 1991; Herrmann et al., 2005). The model has proven effective under various climatic conditions (Kornher and Torssell, 1983; Fagerberg and Nyman, 1994, 1995). For over 10 years, the model has been used by the official agricultural advisory boards – in cooperation with the German Weather Service – in several regions in Germany for predicting the optimum harvest date of primary growths of grass swards. If adapted to maize, it may meet all requirements of a simulation model, predicting yield and quality simultaneously and with satisfactory precision.

The main objectives of the present study therefore were: (i) to modify FOPROQ from a forage grass to a forage maize model. We characterised maize forage quality by dry matter and starch content. (ii) To calibrate and validate the modified FOPROQ maize version for six cultivars covering a wide range of maturity classes (early to mid-late) and maturity types based on multi-year, multi-site field experiments. Finally, we aimed (iii) to quantify and analyse, by means of a 25-year simulation, the climatic risk of maize production for regions in Northern Germany, which are marginal for reliable forage maize production.

2. Materials and methods

2.1. Model description

FOPROQ is one of only a few models covering yield and quality aspects of forage crop simulation. Growth calculations are based on weather data as well as on plant and soil characteristics. FOPROQ consists of two sub-modules, one for DM yield (Fig. 1, left) and one for forage quality (Fig. 1, right). The model requires as input daily data average air temperature (°C), precipitation (mm), potential rates of evapotranspiration (mm), and incident global radiation (MJ m^{-2} d^{-1}). Maize DM accumulation starts when a threshold GDD value is reached. Growth is simulated in daily time steps as a function of the current amount of biomass and the relative growth rate, which is a product of the growth potential of the young crop and an age index, the latter describes the impact of crop

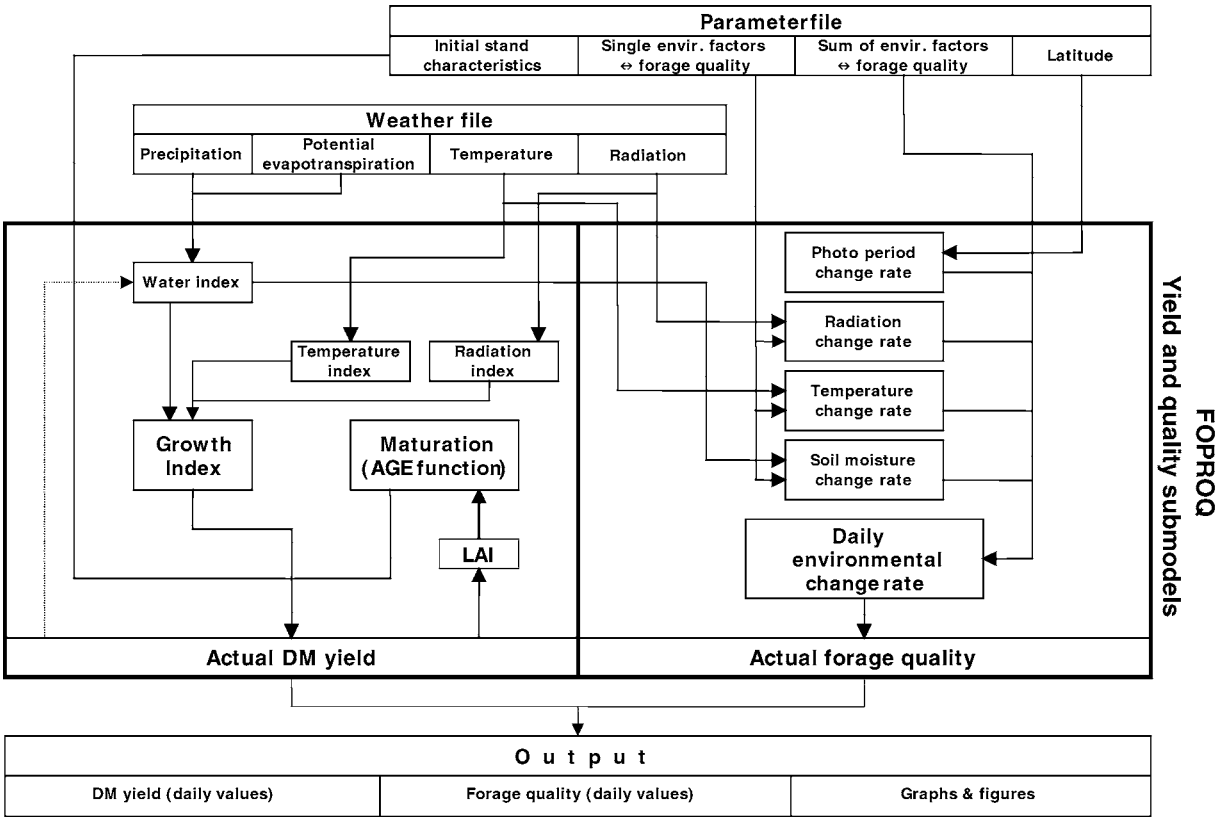

Fig. 1. Flow chart representing the main processes of the FOPROQ model.

ageing on growth potential. A growth index, given by the product of the indices of temperature, radiation and plant-available soil water, represents the weather influence on growth. This 'environmental factor' reduces the potential growth rate to an actual rate. Genotypic differences in growth potential are accounted for by the initial amount of biomass, the relative growth rate at the onset of the vegetation period, and the parameters of the age index. For further details on the model algorithms see Appendix A.

The sub-module for quality prediction assumes the existence of different levels of quality over the entire growing period, with changes from one level to another occurring gradually. The present model, however, allows only for two such levels. The levels and changes in quality depend on input variables reflecting genetics, environment, and management. Environmental factors (as temperature, photoperiod, radiation, and plant-available soil water) are converted into corresponding change rates based on corresponding exponential or negative exponential functions. Daily environmental change rates are obtained as the product of the rates for temperature, photoperiod, radiation, and plant-available soil water, and are related to forage quality change by a threshold–response function. Each simulation run generates as output a set of files containing data such as accumulated biomass and forage quality on a daily basis. We deactivated the photoperiod change function, since it was developed for long-day plants, accelerating quality change for day lengths exceeding a threshold value. Concerning the DM content we assumed an increase for higher temperature, higher radiation intensity, and under drought conditions. With respect to starch accumulation higher temperature and irradiation were supposed to accelerate quality change, whereas drought conditions were assumed to reduce the assimilate flux into the kernels (Setter et al., 2001).

Model calibration was conducted by an integrated parameter optimisation module, which minimises the deviation between simulated and experimental data in terms of the sum of squared residuals. Model parameters were optimised for each cultivar separately. The goodness of fit was then assessed by three methods, the root mean square error (RMSE), the coefficient of determination, and by slope and intercept of the linear regression of predicted versus observed data.

2.2. Calibration data set

The modified FOPROQ model was calibrated against data on aboveground DM yield, DM content, and whole crop starch content. The data were collected in a field experiment conducted at four sites in Germany over three subsequent years (Table 1). Maize was sown between late April and early May in rows 0.75 m apart. The final plant density was 7–10 plants m^{-2} depending on site and variety. The amount of applied N was adjusted to local growth conditions in order to allow maximum production, but was limited to a maximum of 150 kg N ha^{-1}. Plant protection, phosphorus and potassium fertilisation was applied according to the codes of 'Good Agricultural Practice in Plant Protection and Fertilisation'. The experiment was designed as a split-plot with 2–3 replicates. The main plots were six cultivars ('Janna', 'Justina', 'Elita', 'Noveta', 'Calimera', and 'Benicia') covering a wide range of maturity classes relevant for Germany from early to mid-late, including different maturity types (normal, dry-down and stay-green; low to high harvest index). Unfortunately, not all cultivars could be tested on all sites in each year. Missing are the following six cultivar/site/year combinations: Janna/Freiburg/1998, Janna/Freiburg/2000, Justina/Freiburg/1998, Justina/Buxtehude/1998, Benicia/Kiel/1999, and Benicia/Kiel/2000. The sampling dates were assigned to the sub-plots, allowing for a maximum of 12 dates. Samples were taken at 2-weeks intervals starting 3–4 weeks after emergence. The sampling schedule included the recording of crop phenology on each date, the occurrence of key growth stages, e.g. tasseling and silking, and the collection of plant samples for yield and quality determination. Twenty consecutive plants, randomly assigned to a row section bordered by un-harvested rows, were clipped, weighed and chopped as whole crop on each sampling date. A representative sub-sample was oven-dried at 65 °C to constant weight for forage quality analysis. Another sub-sample was dried at 105 °C to determine DM content. After drying, the 65 °C samples were ground in a Tecator mill (Foss Tecator AB, Höganäs, Sweden) in a two-step procedure: first to pass through a 4-mm sieve, and subsequently through a 1-mm sieve. Starch content was determined using near infrared reflectance spectroscopy (NIRS), and starch content of the NIRS calibration and validation subsets was measured polarimetrically as

Table 1
Characteristics of the experimental field sites used for model calibration and validation

Location/year	Geographic coordinates	Altitude (m a.s.l.)	Sowing date	Soil
Calibration				
Buxtehude, 1998	53°25N 9°37E	22	24/4/1998	Loamy sand
Buxtehude, 1999	53°25N 9°37E	22	10/5/1999	Loamy sand
Buxtehude, 2000	53°25N 9°37E	22	18/4/2000	Loamy sand
Braunschweig, 1999	52°18N 10°27E	75	28/4/1999	Loamy sand
Braunschweig, 2000	52°18N 10°27E	75	3/5/2000	Loamy sand
Freiburg, 1998	47°58N 7°41E	250	11/5/1998	Silty loam
Freiburg, 2000	47°56N 7°41E	250	20/4/2000	Silty loam
Kiel, 1999	54°25N 9°56E	32	2/5/1999	Sandy loam
Kiel, 2000	54°18N 9°58E	32	29/4/2000	Sandy loam
Validation				
Bad Hersfeld, 2001	50°50N 9°40E	200	24/4/2001	Loamy sand
Bad Hersfeld, 2002	50°50N 9°40E	200	10/5/2002	Loamy sand
Bernburg, 2001	51°49N 11°42E	80	8/5/2001	Sandy loam
Bernburg, 2002	51°49N 11°42E	80	13/5/2002	Sandy loam

described in Naumann et al. (1997). Meteorological data were recorded either on-site or provided by the German Weather Service for the nearest weather station.

2.3. Validation field experiment

The model was validated with data collected in an independent project on two sites (Bad Hersfeld and Bernburg) over two years (2001 and 2002), see Table 1. The validation of model performance was restricted to the variety 'Benicia', the only variety, which had been tested in both field experiments. The validation experiment was in a split-plot design with two replications, where eight cultivars were assigned to the main plots and the sampling dates to the sub-plots. Crop management (i.e. sowing, fertilisation schedule, and plant protection measures) in the validation experiment was comparable to that in the calibration experiment. At each of seven sampling dates (two before and five after silking) 10 plants were taken. Processing of the samples, i.e. chopping, drying, determination of DM content, was the same as in the calibration experiment. The NIRS technique was used to predict starch content. Wet laboratory analyses of the calibration and validation sets for starch content were conducted according to Naumann et al. (1997). Weather data were collected on-site and supplemented by data of the German Weather Service for the nearest station available.

2.4. Design of the simulation study

With the introduction of earlier-maturing, lodging-resistant cultivars during the last decades silage maize production expanded progressively in Northern Germany and Denmark despite marginal climatic conditions for maize production. Applying the calibrated model, a long-term simulation study permitted quantification of the weather-related variability of maize performance in terms of DM yield and of forage quality, i.e. contents of DM and starch. The simulation study was conducted for the Kiel site, which represents the marginal areas of Northern Germany and Southern Denmark, where unfavourable weather conditions, especially the combination of low temperature and irradiation, tend to constrain silage maize maturity. Twenty-five years of daily

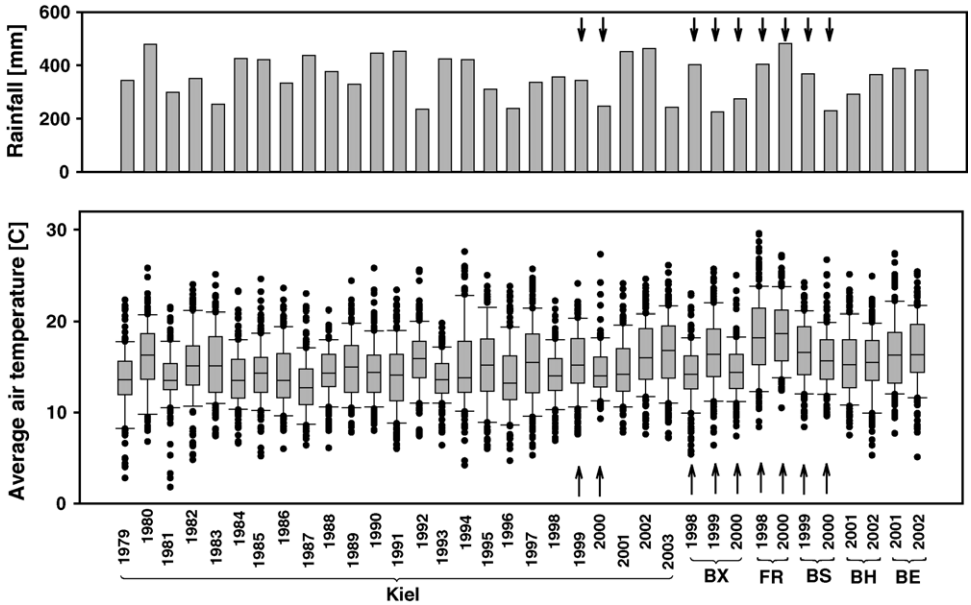

Fig. 2. Climatic conditions given as distribution of daily mean air temperature (°C) and precipitation sum (mm) for the vegetation periods (1 May–10 October) of the calibration (indicated by arrows) and validation (Bad Hersfeld and Bernburg sites in 2001 and 2002) data sets, and for the 25-year simulation of the Kiel site. The boxes give the 25 and 75% percentiles and the median; the whisker caps indicate the 10 and 90% percentiles. Sites are abbreviated as follows—BE: Bernburg; BH: Bad Hersfeld; BS: Braunschweig; BX: Buxtehude; FR: Freiburg.

weather records (1979–2003), kindly provided by the German Weather Service, were used to run the simulation. The parameter sets obtained by the preceding model calibration were assumed suitable for a long-term simulation, since the calibration data span a wide range of weather conditions, as indicated by the distribution of average air temperature and accumulated amount of rainfall in Fig. 2, right hand side.

The simulation study was conducted using the module FOSIM, which allows to run FOPROQ with a pre-specified harvest strategy, controlling the harvest time by setting target values concerning harvest date, DM yield, and/or forage quality. As soon as one of the target values is reached, the simulation run is terminated, and the results are given as daily values of biomass, growth rates, forage quality, and model-specific environmental indices. For silage maize, we set the harvest time for all varieties in the study at a whole crop DM content of 320 g DM kg^{-1} fresh weight. For unfavourable climatic conditions with delayed maturity we set 10 October as the latest possible harvest date if silage maturity was not reached until then. A risk assessment for forage maize production was performed for the 25-year period, analysing the frequency distributions of harvest date, of DM yield, of DM content, and of starch content for three hybrids ('Janna', 'Justina', and 'Noveta'), which cover the range of maturity groups recommended for Northern Germany.

3. Results

3.1. Model calibration

The FOPROQ model could be applied to forage maize without modification of any model algorithm. It comprises a total of 54 parameters, 28 originating from the growth part and 26 from the quality part. Out of the 28 growth-related parameters it was possible to assume 23 as identical for all cultivar-site combinations. For three parameters cultivar-specific values had to be assigned. Two additional growth parameters for calculation of soil water balance needed site-specific value assignments. Similarly, most of the 26 quality-related parameters were cultivar- and site-independent. Optimisation of DM content and of starch content resulted in six and eight cultivar-specific parameters, respectively (Table 2).

Table 2
Results of parameter optimisation of the FOPROQ growth and quality sub-models

	Janna	Justina	Elita	Noveta	Calimera	Benicia
Growth model						
W_0	0.960	0.965	0.990	1.020	0.990	0.700
r	0.210	0.217	0.199	0.199	0.202	0.214
a	2.32	1.90	1.95	2.00	1.75	1.57
Quality model: DM content						
t_c	0.346	0.302	0.300	0.300	0.360	0.330
T_{th}	6.0	6.0	6.0	6.0	6.0	6.0
r_c	0.191	0.168	0.164	0.164	0.164	0.124
R_{th}	3.0	3.0	3.0	3.0	3.0	3.0
m_c	5.0	5.0	5.0	5.0	5.0	5.0
Q_n	11.70	11.55	11.60	11.10	11.35	11.15
Q_x	73.40	98.90	133.60	132.25	150.10	110.00
v	105.34	114.52	124.2	133.54	139.68	113.00
c	5.15	4.44	4.52	3.93	4.55	4.69
Quality model: starch content						
t_c	0.107	0.169	0.234	0.261	0.327	0.240
T_{th}	6.0	6.0	6.0	7.0	7.0	8.0
r_c	1.017	0.808	0.601	0.418	0.216	0.269
R_{th}	3.0	3.0	3.0	3.0	3.0	3.0
m_c	0.30	0.30	0.70	0.40	0.50	0.30
Q_n	0.21	0.21	0.52	0.27	0.52	0.25
Q_x	38.47	33.70	39.57	39.90	32.35	32.28
v	81.67	97.26	113.44	109.11	108.17	97.41
c	11.82	12.13	12.47	10.47	17.84	17.98

For further details with respect to model parameters see Appendix A.

Model performance was assessed by R^2 and RMSE, and by the regression of simulated versus observed data. Model fit was determined for each cultivar over all calibration sites (Table 3). In order to detect potential systematic deviations caused by environmental conditions or crop management, model fit statistics for each site were calculated over all cultivars (Table 4).

3.1.1. Dry matter accumulation

The three cultivar-specific growth parameters in the model are the initial biomass W_0, the young crop's relative growth rate r, and the shape parameter a of the age function A (Appendix A, Eqs. (A.1) and (A.2)). Model calibration resulted for parameters W_0 and r in uncorrelated estimates, which showed no relationship to their maturity classes. Differences among varieties with respect to the time course of biomass accumulation

Table 3
Results of model calibration for dry matter (DM) yield (g DM m^{-2}), DM content (g DM kg^{-1} fresh weight), and starch content (g kg^{-1} DM). Model fit is given for each hybrid separately as number of observations (n), coefficient of determination (R^2), root mean square error (RMSE), and the intercept (a) and slope (b) of the calculated versus observed regression equation

Hybrid	n	R^2	RMSE	a	b
DM yield (g DM m^{-2})					
Janna	69	0.95	127.9	0.21	0.999
Justina	66	0.95	147.6	0.22	0.996
Elita	84	0.94	146.1	−0.03	0.998
Noveta	85	0.94	148.3	0.67	1.000
Calimera	84	0.95	149.9	0.19	0.995
Benicia	65	0.92	209.6	−0.24	0.998
DM content (g DM kg^{-1} fresh weight)					
Janna	70	0.93	27.4	−0.44	0.999
Justina	66	0.94	27.3	−0.06	1.000
Elita	77	0.94	27.2	0.10	1.000
Noveta	77	0.93	26.6	−0.54	1.000
Calimera	77	0.92	25.6	0.11	1.000
Benicia	57	0.93	27.1	−0.05	1.000
Starch content (g kg^{-1} DM)					
Janna	57	0.94	28.79	1.80	0.990
Justina	53	0.94	30.38	−0.01	1.000
Elita	65	0.91	34.94	−1.15	0.990
Noveta	71	0.88	45.48	−0.85	1.017
Calimera	72	0.93	29.41	−0.86	1.003
Benicia	45	0.94	28.38	−3.00	0.951

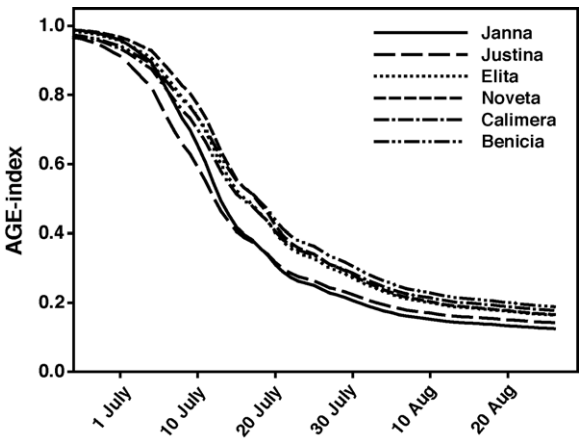

Fig. 3. Time course of the age function depending on the cultivar involved, exemplified for the Buxtehude site in 1999. Silking was observed between 14 July and 24 July.

required cultivar-specific parameter assignments of the parameter a. As exemplified in Fig. 3 for Buxtehude 1999, the age functions of 'Janna' and 'Justina' differ considerably from the later-maturing cultivars. The comparison of calculated and observed data in Table 3 demonstrate that biomass accumulation was reasonably well estimated for all cultivars. Prediction errors are in an acceptable range with RMSE values accounting for about 10% of the DM yield at silage maturity. Table 4 displays indicators of model fit for each experimental site. They also support the accuracy of the model predictions, except for the Freiburg 1998 data. This could be traced back to a discrepancy of 45 days (Table 5) between simulated sowing date and actual one, resulting in an overestimation of biomass production in the earlier growth stages, as exemplified in Fig. 4 for 'Elita' and 'Noveta'.

3.1.2. Dry matter content

The six cultivar-specific DM content parameters were t_c, r_c, Q_n, Q_x, c, and v (Table 2). Model optimisation confirmed the major impact of temperature on the increase of whole crop DM content. The model's temperature and radiation change functions contain two constants describing the shape of the function (t_c and r_c) and representing a threshold value (T_{th} and R_{th}, Appendix A, Eqs. (A.4) and (A.5)). Estimation of t_c and r_c revealed differences among cultivars and had to be set cultivar-specifically. The threshold temperature T_{th} and irradiation R_{th}, in contrast, turned out to be essentially cultivar-independent. This is remarkable since Bloc and Gouet (1977) and Derieux and Bonhomme (1990) found maize inbred lines to have a characteristic cultivar-specific base temperature, ranging between 0 and 6 °C. The parameter m_c in the model's moisture change function (Appendix A, Eq. (A.6)), could be assigned uniformly for all cultivars. For situations of water deficiency the function increases DM content change rates. In general, however, soil water availability was of minor importance. Estimations for the threshold–response function, which quantifies the relationship between the accumulated daily environmental change rates and DM content (Appendix A, Eq. (A.8)), showed substantial differences among varieties for parameters Q_x, v, and c. Values for Q_x, the maximum DM content at maturity have no physiological meaning, but were caused by the experimental

Table 4
Indicators of model fit for dry matter (DM) yield (g DM m^{-2}), DM content (g kg^{-1} fresh weight), and starch content (g kg^{-1} DM), given as number of observations (n), coefficient of determination (R^2), root mean square error (RMSE), and the intercept (a) and slope (b) of the calculated versus observed regression equation for each site–year combination separately

	Buxtehude 1998	Buxtehude 1999	Buxtehude 2000	Freiburg 1998	Freiburg 2000	Kiel 1999	Kiel 2000	Braunschweig 1999	Braunschweig 2000
DM yield (g DM m^{-2})									
n	55	53	66	32	35	45	55	59	53
R^2	0.98	0.95	0.97	0.93	0.96	0.98	0.98	0.92	0.97
RMSE	86.3	194.6	118.8	304.9	137.5	104.8	106.9	184.0	127.7
a	36.16	−164.07	57.14	−473.49	−43.42	29.03	67.61	109.40	−107.19
b	0.96	1.22	0.94	1.21	1.04	1.03	0.99	0.89	1.11
DM content (g DM kg^{-1} FM)									
n	55	54	66	n.a.	35	45	55	60	54
R^2	0.95	0.97	0.92	n.a.	0.98	0.86	0.92	0.96	0.98
RMSE	24.3	23.9	31.7	n.a.	20.3	29.1	22.9	35.8	17.9
a	−1.11	0.42	−0.54	n.a.	1.18	3.32	0.08	2.29	−2.21
b	0.96	0.91	1.11	n.a.	0.94	0.75	1.04	1.00	1.06
Starch content (g kg^{-1} FM)									
n	54	53	53	24	27	44	54	n.a.	54
R^2	0.94	0.92	0.95	0.83	0.98	0.98	0.96	n.a.	0.93
RMSE	32.1	36.6	28.8	57.6	24.3	13.0	34.9	n.a.	38.9
a	0.46	−0.84	0.76	−0.48	−0.05	−0.44	0.48	n.a.	−1.08
b	0.80	1.01	1.06	0.87	0.91	1.01	1.18	n.a.	1.11

n.a.: not available.

design since the maize crop was harvested before physiological maturity. Parameters c and v, which control the inclination and inflexion point of the threshold–response function, are better representations of the cultivar-specific maturity, as exemplified in Fig. 5 for Braunschweig 2000.

Simulated and observed DM concentrations generally agreed well for all cultivars, explaining 92–94% of the observed variation (Table 3). RMSE errors ranged between 25.6 and 27.4 g DM kg^{-1} fresh weight. Some minor discrepancies became apparent in 1999 for the sites of Kiel and Braunschweig, and for Buxtehude 2000 (Table 4). For Kiel, the last sampling dates were overestimated, whereas for Braunschweig and Buxtehude DM concentrations were somewhat underestimated.

Table 5
Comparison of simulated and observed sowing dates, given as calendar days, of the site–year combinations included in model optimisation

Site–year	Observed sowing date	Simulated sowing date
Braunschweig, 1999	118	116
Braunschweig, 2000	124	118
Buxtehude, 1998	114	118
Buxtehude, 1999	130	125
Buxtehude, 2000	109	121
Freiburg, 1998	131	86
Freiburg, 2000	111	95
Kiel, 1999	122	131
Kiel, 2000	120	126

3.1.3. Starch content

The whole crop starch content does not increase until the onset of the effective grain fill period. We decided nevertheless to start at sowing, because drought or nutrient stress during the vegetative phase can delay tassel initiation and silking (Abrecht and Carberry, 1993; Singh and Wilkens, 2000). The cultivar-specific starch parameters are given in Table 2. Among the climatic factors considered, temperature proved to be most important, with increasing temperatures accelerating starch accumulation. In contrast to the DM content optimisation, model fit was improved when cultivar-specific values were assigned to the temperature threshold T_{th} of the temperature change function (Table 2; Eq. (A.4), Appendix A). For the radiation change function, only

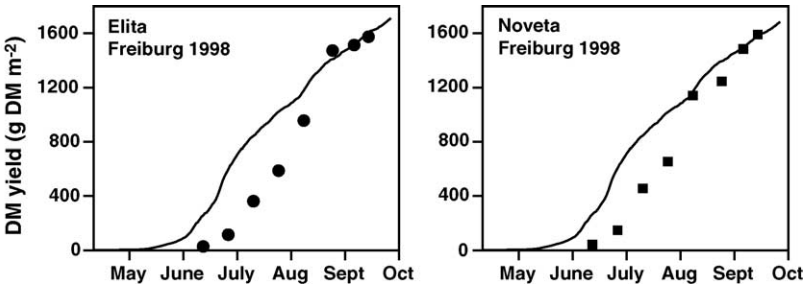

Fig. 4. Estimated (lines) and observed (symbols) dry matter (DM) yields (g DM m^{-2}) of the 1998 growing season at Freiburg, displayed for the cultivars 'Elita' (left figure) and 'Noveta' (right figure).

parameter r_c revealed differences among cultivars, while the threshold R_{th} was estimated uniformly. Water shortage was assumed to slow down starch accumulation (Appendix A, Eq. (A.7)). The corresponding parameter m_c was estimated cultivar-specific, and indicated a higher susceptibility of 'Elita'. As for DM content, parameters Q_x, v, and c estimated for the starch threshold–response function differed among cultivars. Fig. 5 may exemplify that starch accumulation began early for 'Janna' and 'Justina', while it was delayed for 'Calimera' and 'Benicia'. Substantially higher c values for 'Calimera' and 'Benicia' suggest a stronger starch increase. Higher temperature and irradiation requirements together with unfavourable climatic conditions in late summer, however, slowed the starch accumulation of both cultivars down.

Agreement between simulated and measured starch concentrations was satisfactory for all cultivars with R^2 values above 0.90, regression lines of simulated versus observed data close to the bisecting line, and RMSE values around 10% of the final yield and quality values. Only 'Noveta' showed a somewhat higher variation, as indicated by a RMSE value of 45.48 g kg^{-1} DM (Table 3). The satisfactory model fit is confirmed by the indicators calculated for each site–year combination separately (Table 4). Somewhat larger deviations became apparent for Freiburg 1998, mainly due to the late sowing date.

3.2. Model validation

3.2.1. Model performance on a 2-year–2-sites data set

The FOPROQ simulation results were compared with an independent data set on two sites in Bernburg (BE) and Bad Hersfeld (BH) over two years (2001, 2002) for the cultivar 'Benicia'. Model performance is displayed in Fig. 6 for DM yield (left), DM content (middle), and starch content (right).

The DM yield simulations were able to explain 90% of the DM yield. The RMSE value, however, was considerably higher than most corresponding calibration values (Table 4). Larger deviations occurred in

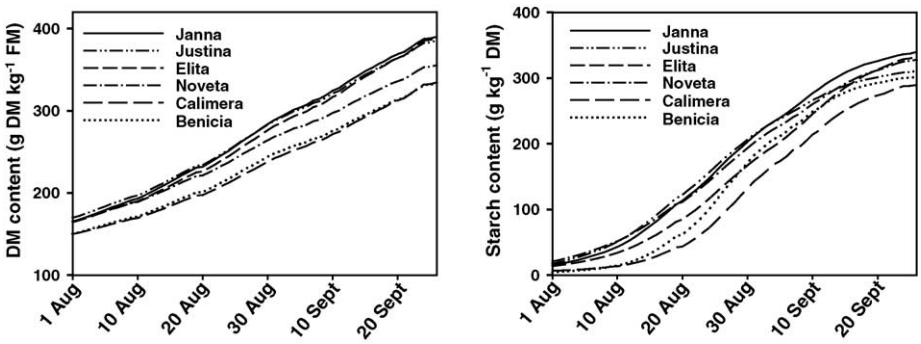

Fig. 5. Simulated hybrid-specific increase of dry matter (DM) content (g DM kg^{-1} fresh matter) and starch content (g kg^{-1} DM), exemplified for the site of Braunschweig in 2000.

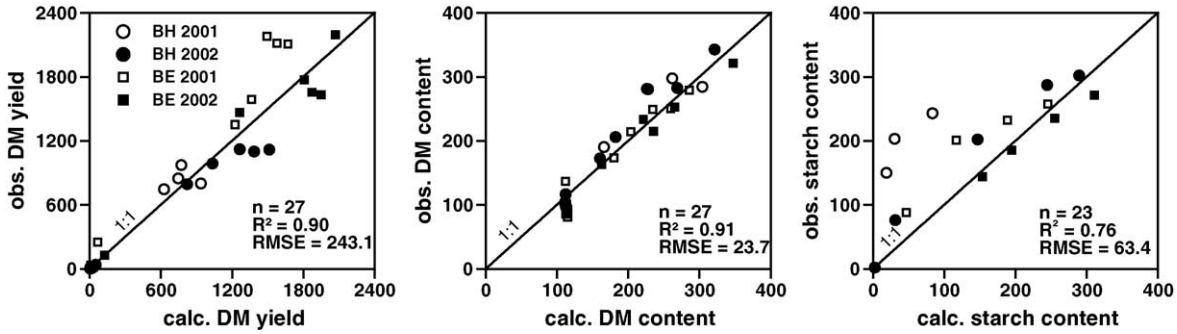

Fig. 6. Results of model validation for dry matter (DM) yield (g DM m^{-2}), DM content (g DM kg^{-1} fresh weight), and starch content (g kg^{-1} DM). Model performance is assessed by the coefficient of determination (R^2) and root mean square error (RMSE). Sites are abbreviated as follows—BH: Bad Hersfeld; BE: Bernburg.

2001 for BE where DM yield was overestimated, smaller deviations were present in BH 2002. These discrepancies most likely resulted from a lack of proper representation of the soil water balance. Specifications of soil conditions, especially of the available field capacity, probably did not have the required accuracy. The severe water shortage, however, which reduced DM yield substantially in BH 2001, was well reflected by the model. The DM content simulations were reasonable with a coefficient of determination of 0.91 and a RMSE value of less than 10% of DM content at silage maturity. Apart from some minor deviations, the model was able to reflect the whole range from low DM contents of the young crop up to 30–35% at silage maturity. The starch content simulations showed a notably lower R^2 of 0.76, mainly caused by an underestimation of drought impact on starch accumulation in BH 2001. The validation results suggest that modelling of starch content under severe drought conditions deserves further improvement. For normal conditions, however, model performance was satisfactory.

3.2.2. Model performance on a 25-year regional data set

Since one main purpose of our study was to assess the climatic risks attached to silage maize production in the marginal regions of Northern Germany, we conducted another test to examine model performance. Most German dairy farmers routinely make use of the services of an existing network of private and official consultancy agencies to conduct a forage quality analysis of their silages. We used the available data on maize silage quality, originating from dairy farms in Schleswig-Holstein (Northern Germany) and compared them with data obtained in the 25-year simulation study. Simulations are shown for 'Justina', which resembles best the maturity type commonly grown in that region.

The DM content data, displayed in Fig. 7 (middle), indicate a generally good model agreement between actual and simulated values. For unknown reasons the correspondence is less close in the first decade. The model was able to reflect the effect of low temperatures, e.g. for 1987, and the impact of water deficiency conditions, as most pronounced in 2003. For starch content (Fig. 7, bottom), observed data unfortunately have been available only since 1987. Simulated and observed values are in slightly less good agreement compared to the DM content data. This may in part be attributed to the cultivar-specific starch accumulation potential, which differed between 'Justina' and the cultivars grown on the farms. Severe water shortage is handled poorly, causing larger underestimations in 1992, 1996, and 2003. Some limitations apply to our comparisons, because the DM and starch content of the simulated green crop and the observed silage data are not fully compatible (Savoie et al., 2002). Furthermore, the observations represent average values of a multitude of cultivars, which will have changed over years. In summary, however, the simulations reproduce the course of the forage maize production in Northern Germany fairly well.

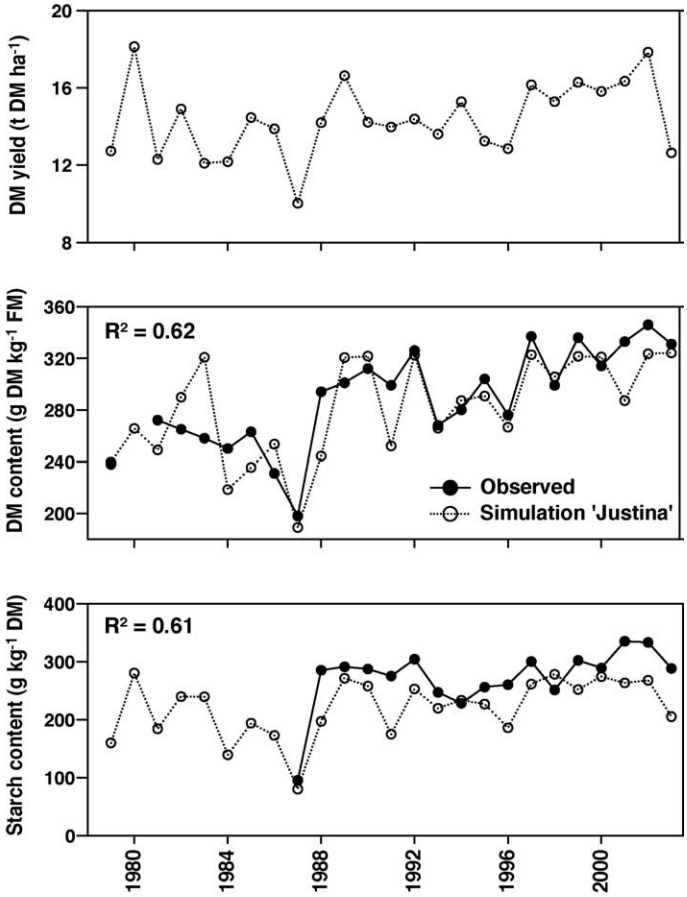

Fig. 7. Validation of the 25-year simulation (1979–2003) for concentrations of dry matter (DM) and starch. Calculated values represent the simulations for the hybrid 'Justina'. Observed data show means for DM and starch concentration of maize silages produced in the state of Schleswig-Holstein, Northern Germany during the simulation period. Data are given as means, based on 60–1426 observations, depending on the year. Actual DM yield data unfortunately were not available. The data were kindly provided by the Agricultural Chamber of Schleswig-Holstein.

3.3. Simulation-based climatic risk assessment

In order to study the climatic risk attached to silage maize production in the marginal regions of Northern Germany, the distributions of the obtained 25-year simulation results (1979–2003) of harvest date, DM yield, DM content, and starch content were analysed. We characterised the location of the distributions by their mean and median, because most distributions were left skewed. The dispersion of the distributions was quantified by the coefficient of variation (cv) and by the excess kurtosis, where positive values indicate a peaked distribution and negative values indicate a flat distribution, i.e. a risk of high instability. In Table 6 all analysed statistics are listed for each of the three varieties 'Janna', 'Justina', and 'Noveta', and Fig. 8 displays the corresponding cumulative frequency curves.

The harvest date in the simulations was defined to be reached either when the target DM content of 320 g DM kg^{-1} fresh weight was achieved, or on 10 October (calendar day 283) the simulation run was terminated automatically. The mean of the 25 simulated harvest dates was close to 283 for all three varieties, with relatively narrow variation as indicated by the small cv values between 2.55 and 3.28, and by positive kurtosis values (Table 6). In 64% for 'Janna' and 'Justina', and in 76% for 'Noveta' the simulation failed at all to reach the target DM content. The impact of early autumn frosts was not considered in the model

Table 6
Results of the simulation study for the 25-year period from 1979 to 2003, given as mean, median, coefficient of variation (cv) in percent, and excess kurtosis for harvest date (calendar day), dry matter (DM) yield (t DM ha^{-1}), DM content (g DM kg^{-1} fresh weight), and starch content (g kg^{-1} DM)

	Janna	Justina	Noveta
Harvest date (calendar day)			
Mean (median)	277 (283)	278 (283)	280 (283)
cv	3.28	3.17	2.55
Excess kurtosis	1.39	2.35	5.30
DM yield (t DM ha^{-1})			
Mean (median)	12.6 (12.5)	14.4 (14.2)	12.8 (12.7)
cv	11.04	13.45	14.56
Excess kurtosis	−0.05	−0.09	−0.21
DM content (g DM kg^{-1} fresh weight)			
Mean (median)	286.3 (293.6)	281.5 (287.2)	271.0 (270.9)
cv	12.80	13.92	15.22
Excess kurtosis	−0.20	−0.56	−0.96
Starch content (g kg^{-1} DM)			
Mean (median)	203.8 (227.4)	220.3 (233.4)	210.7 (204.4)
cv	31.78	22.86	29.29
Excess kurtosis	−0.22	0.85	−0.54

calculations. The occurrence of early frosts is unlikely due to the vicinity of the North and the Baltic Sea, which provide a heat reservoir. If we define a killing frost by a daily mean air temperature below −1 °C, then for none of the 25 years such an instance was recorded.

The DM yield simulation showed substantial differences among cultivars for all statistics calculated. Mean DM yield was highest for the early hybrid 'Justina'. Yield stability declined with decreasing earliness, as indicated by cv values ranging from 11.04 ('Janna') to 14.56 ('Noveta'), and kurtosis values between −0.05 ('Janna') and −0.21 ('Noveta'). The extremely low DM yields in 1987 (Fig. 7) were mainly due to low temperature, whereas the considerably reduced yields in 2003 were caused by water shortage. The yield outliers in the upper range of the distributions as those in 1980 and 2002 are best explained by high temperatures and above-average precipitation, which provided favourable growth conditions.

While the average DM content between early 'Janna' and mid-early 'Noveta' differs by only 15 g DM kg^{-1} fresh weight, the DM content variation

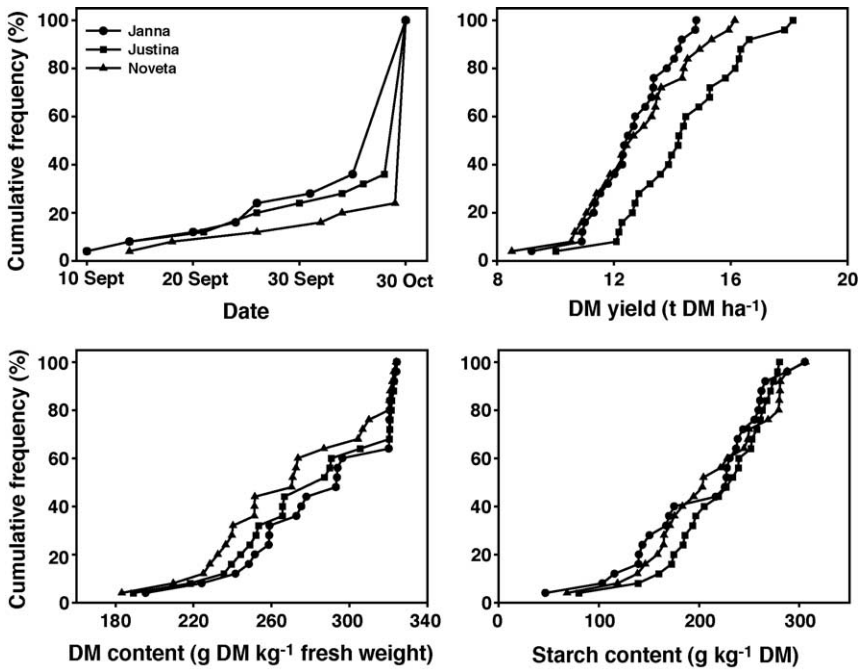

Fig. 8. Results of the 25-year simulation study given as cumulative frequency distribution of harvest day (day of year), dry matter (DM) yield (t DM ha^{-1}), DM content at harvest (g DM kg^{-1} fresh weight), and starch concentration at harvest (g kg^{-1} DM) for the hybrids involved ('Janna', 'Elita', and 'Noveta').

between these two varieties is considerably distinct as expressed by the corresponding kurtosis and cv values. With increasing earliness the risk of low DM contents reduces. As already mentioned, the silage maturity target (320 g DM kg^{-1} fresh weight) was missed in 64% for 'Janna' and 'Justina' and in 76% for 'Noveta'. This, however, does not imply a complete failure of silage maize production since the simulated DM content was close to the target value for a number of years. If we regard 290 g DM kg^{-1} as a lower limit for a successful ensiling under avoidance of effluent, the likelihood of failure declines to 44% ('Janna'), 52% ('Justina'), and 64% ('Noveta'), respectively.

For starch content intersecting cumulative frequency curves (Fig. 8) suggest a less pronounced differentiation among cultivars, especially for values above 220 g kg^{-1} DM. With respect to starch content in contrast to DM content, the mean was lowest for the early cultivar 'Janna', while the median was lowest for 'Noveta' (Table 6). Also the variation of starch content did not reflect the maturity pattern observed for DM content. A small variation (positive kurtosis, small cv) exhibited by 'Justina' suggests a lower sensitivity to environmental conditions compared to 'Janna' and 'Noveta'. With cv values between 22.9 and 31.8 variability of starch content was generally higher than of DM content (Table 6).

4. Discussion

4.1. Need for model refinement

Model validation, especially for starch content, revealed some model weakness when the crop was affected by severe water deficiency, as for Bad Hersfeld in 2001. A lack of strong water shortage conditions in the calibration data set may in part account for this model shortcoming. Furthermore, the effect of water shortage on starch deposition depends on the particular developmental stage. The current FOPROQ version, however, does not allow a differentiation of the response functions of temperature, radiation and soil water with respect to developmental phases.

Future model refinement first of all requires a broader database, which includes more sites vulnerable to water deficit in order to increase the robustness of model parameters. Performance might be further enhanced by model modifications such as (i) a differentiation of the environmental response functions with respect to developmental stages; and (ii) the use of soil instead of air temperature for the simulation of early growth stages, which will be discussed below in more detail. These modifications, however, involve the introduction of additional model parameters, which in turn may reduce the model robustness substantially.

4.1.1. Development-stage specific response functions

The revealed weaknesses of the FOPROQ model with respect to starch content prediction may be due to a lack of a process-oriented structure. Individual processes are not described in detail, but only broadly explained by climatic conditions as the driving variables of crop growth and development. While the low complexity of the model and its sparse input data requirement are advantageous for practical application, the costs with respect to accuracy may be too high. If crop sensitivity to drought stress varies substantially among developmental phases, model performance might be enhanced by distinguishing different developmental stages, for instance vegetative and reproductive stages. This partitioning would allow to estimate response functions separately for each developmental period in order to quantify the impact of environmental conditions more accurately. Such approach is supported by Stewart et al. (1998), who found substantial differences in the phenological temperature response between the vegetative and reproductive growth stages. Similarly, Birch et al. (1998) reported base temperature for crop ontogeny to differ between the pre- and post-silking phase. However, conflicting to this are results, e.g. by Yan and Hunt (1999), who suggest that in most maize cultivars the cardinal temperatures for growth and development processes coincide essentially. Following this line, growth models like CERES Maize and CropSyst (Jones and Kiniry, 1986; Stöckle et al., 1994) use common base temperature and optimum temperature for crop phenology over the whole growth cycle.

4.1.2. Soil temperature versus air temperature

The thermal time approach is successfully used in many models to describe the temperature impact on

the rate of crop development or growth. While temperature–response functions vary considerably (Hunt et al., 2003), they commonly use air temperature as thermal parameter. Several studies indicate that soil temperatures provide greater accuracy for predicting crop phenology and growth (Cutforth and Shaykewich, 1989; Jamieson et al., 1995). CERES Maize, for instance, uses soil temperature for growth stages up to the tenth leaf or tassel initiation, when the shoot apex still is near soil surface (Jones and Kiniry, 1986). Contrary results were reported by McMaster and Wilhelm (1998), who could not find any significant improvement of using soil instead of air temperature for predicting phenological stages of winter wheat. The current version of FOPROQ provides no feature that allows to switch from soil to air temperature depending on crop phenology. A modified version is currently under development, and will include such an option for testing if model accuracy is enhanced by switching to soil temperatures for early growth stages.

4.2. Climate change and risk of silage maize production

The 25-year simulation study showed that the reliability of silage maize production in Northern Germany is severely constrained by environmental conditions, especially by low temperatures. Our results are in good agreement with the study of Wilson et al. (1994) on climatic risk of maize production in the South Island of New Zealand. They found similar crop failure probabilities for sites with comparable environmental conditions.

Long-term studies on climate have demonstrated that mean air temperature has risen over the last decades, and further temperature increase is projected for the 21st century (IPCC, 2001). A strong change for annual average air temperature is also present in the weather records used in our simulation study. During the last 10 years the mean air temperature of the vegetation period (1 May–10 October) increased by 0.14 °C a^{-1}, whereas precipitation remained on the same level at the Kiel site (data not presented). If we therefore consider only the simulation output of the last 10 years, the climatic risk decreases remarkably, with earlier harvest dates, higher means for DM yield, DM content, and starch content, and a lower variability of yield and forage quality (Table 7). For a target DM content of 320 g DM kg^{-1} fresh weight, the risks are reduced to 40, 50, and 60% for 'Janna', 'Justina', and 'Noveta', respectively. For the lower target of 290 g DM kg^{-1} the crop failure probabilities decrease even more, namely to 10, 30, and 50%, respectively.

Table 7
Results of simulation study for the 10-year period from 1994 to 2003, given as mean, median, coefficient of variation (cv) in percent, and excess kurtosis for harvest date (calendar day), DM yield (t DM ha^{-1}, DM content (g DM kg^{-1} fresh weight), and starch content (g kg^{-1} DM)

	Janna	Justina	Noveta
Harvest date (calendar day)			
Mean (median)	274 (281)	274 (282)	277 (283)
cv	4.39	4.35	3.73
Excess kurtosis	−1.13	−0.72	−0.02
DM yield (t DM ha^{-1})			
Mean (median)	13.2 (13.4)	15.2 (15.5)	13.6 (14.0)
cv	9.25	11.35	12.07
Excess kurtosis	−1.11	−0.87	−1.19
DM content (g DM kg^{-1} fresh weight)			
Mean (median)	308.8 (320.5)	305.0 (313.3)	294.7 (296.9)
cv	5.65	6.79	9.24
Excess kurtosis	−0.65	−0.89	−1.68
Starch content (g kg^{-1} DM)			
Mean (median)	237.4 (240.4)	245.7 (256.5)	239.0 (247.3)
cv	15.61	12.69	20.07
Excess kurtosis	3.18	−0.31	−1.59

These results suggest that the projected temperature change will favour silage maize production and reduce future climatic risks for the marginal regions of Northern Germany. This is in accordance with the study of Wolf and van Diepen (1994) on climate change impact on silage maize production potential in the European Community. Their climate change scenarios resulted in higher yields for regions, where low temperature but not water supply was the limiting factor.

5. Conclusions

The FOPROQ model was originally developed for forage grasses, but turned out to be applicable to forage maize without modifications of any of its algorithms. The experimentally obtained yield and DM content data were reproduced with acceptable accuracy. The model's simplicity is of great advantage for its practical application. On the other hand, the

lack of a more comprehensive modelling of growth and development processes resulted in some shortcomings, such as a considerable underestimation of starch content under drought conditions. Most likely, model performance can be enhanced by distinguishing crop response to environmental stress factors depending on developmental stage. Additional improvement of the model could be achieved by a broader database with more sites vulnerable to water shortage.

The model proved suitable for analysing and quantifying climatic risks of silage maize production in the marginal regions of Northern Germany, showing a considerable risk of failing to reach silage maturity in these regions. The observed increase of annual mean temperature over the last decade reduced the simulated climatic risk, indicating that the projected climate change might considerably decrease the risk of crop failure in the marginal regions of Northern Germany. The reliable results presented in this study support current efforts to develop FOPROQ into a harvest time prognosis tool for Germany. We are fully aware of the fact that harvest time prognosis is worthless if weather conditions do not allow a timely harvest. But for other weather conditions such a prognosis tool could become extremely helpful to farmers, contractors, and advisors.

Acknowledgements

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Appendix A

A.1. Model algorithms

The following section gives a brief overview of the key model components of FOPROQ, for further details see Kornher and Torssell (1983), Kornher et al. (1991), and Fagerberg and Nyman (1995).

A.1.1. Growth sub-model

FOPROQ model calculations start when a GDD value of 153 (6 °C base temperature) is reached. Since biomass simulations generally show a lag phase, this GDD threshold should be interpreted as sowing date rather than the actual onset of aboveground biomass production. The daily change of biomass can then be estimated as follows:

$$\Delta W = WrAG_{\mathrm{i}}, \tag{A.1}$$

and

$$A = \frac{1}{(1 + (L/L_{50})^{a})}, \tag{A.2}$$

where W, daily change of aboveground biomass per unit ground area (g m^{-2}); r, relative growth rate at onset of growth (g g^{-1} d^{-1}); A, age function of leaf area index L to describe the decrease of relative growth rate with crop ageing, scaled between 0 and 1. Leaf area index L is calculated as a function of biomass (not presented); G_{i}, a growth index summarising the impact of weather conditions on growth rate calculated as a product of the indices of temperature (°C), incident global radiation (MJ m^{-2} d^{-1}), and plant available soil water (mm) scaled between 0 and 1; L_{50}, half maximum LAI; a, constant determining the curvature of the function.

The function relating mean daily temperature (°C) to temperature index is assumed bell-shaped. Optimum, base and maximum temperature were optimised 26, 6, and 42 °C, respectively, for all varieties uniformly. The relationship between incident global radiation R_{s} (MJ m^{-2} d^{-1}) and radiation index R_{i} is described as: $R_{\mathrm{i}} = (1 - \mathrm{e}^{(-kR_{\mathrm{s}}/R_{\mathrm{x}})})/(1 - \mathrm{e}^{-k})$, with R_{x} the radiation at light saturation of the canopy, and k a constant. Plant-available soil water S (mm) is simulated with a single layer budget in relation to precipitation P (mm), and depletion by actual evapotranspiration E_{a}:

$$\Delta S = \begin{cases} P_{\mathrm{t}} - E_{\mathrm{a}} & \text{if} \quad S < S_{\mathrm{c}} \\ 0 & \text{else} \end{cases}, \tag{A.3}$$

with S_{c}, the available field capacity.

At field capacity, E_{a} equals the potential evapotranspiration E_{p}. With decreasing soil water content E_{a} is reduced gradually according to the ratio of S to a soil water threshold S_{th}. Potential evapotranspiration, provided by the weather service, is reduced linearly

as a function of leaf area index L, if L values fall below a threshold value. The ratio of E_a to E_p gives the soil water index W_i, which links the growth part of the model to the quality part.

A.1.2. Quality sub-model

The quality model assumes the existence of two levels of quality. Over the growing period, changes from one level to the other occur gradually. The levels and changes in quality depend on genetical, but also on management and environmental input. Environmental factors like temperature, radiation, and soil water are converted into corresponding change rates based on proper exponential or negative exponential functions.

Temperatures above a temperature threshold T_{th} are assumed to accelerate the change of dry matter (DM) content and starch content:

$$T_r = \begin{cases} 0 & T \le T_{th} \\ 1 - (\exp(-t_c(T - T_{th})) & T > T_{th} \end{cases} \quad (A.4)$$

with T denoting the daily mean temperature (°C), and t_c a temperature constant.

The corresponding function for the impact of irradiation is given as:

$$R_r = \begin{cases} 0 & R_s \le R_{th} \\ 1 - \exp(-r_c(R_s - R_{th})) & R_s > R_{th} \end{cases} \quad (A.5)$$

where r_c is a radiation constant and R_{th} a radiation threshold.

For plant available soil water, a linear function is more appropriate. If soil moisture deficit is assumed to increase quality change, such as for DM content, then soil moisture change rate is defined as:

$$M_r = \begin{cases} 0 & W_i > m_c \\ 1 - \left(\dfrac{1}{m_c/W_i}\right) & \text{else} \end{cases} \quad (A.6)$$

with m_c a water availability constant, and W_i the water index, ranging between 0 and 1.

If water shortage is supposed to reduce quality change rates, such as for starch content, then soil moisture change rate is calculated as:

$$M_r = \begin{cases} \dfrac{1}{(m_c/W_i)} & W_i < m_c \\ 1 & \text{else} \end{cases} \quad (A.7)$$

The daily change rates are combined multiplicatively to give the daily environmental change rate P_r, ranging from 0 to 1. The daily rates are summed to get the accumulated daily rate S_r, and forage quality traits are then modelled as functions of S_r. The relationship between S_r and contents of DM (g (100 g)$^{-1}$ DM) and starch (g (100 g)$^{-1}$ DM) is described by a threshold–response curve of switch-on type:

$$Q(S_r) = Q_n + \left(\frac{(S_r/v)^c}{1 + (S_r/v)^c}\right)(Q_x - Q_n), \quad (A.8)$$

where Q, value of quality trait; Q_n, minimum content at the onset of growth; Q_x, maximum content at maturity; c, determines the inclination of the curve; v, is the value of S_r at the inflexion point of the curve.

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